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Effect of pooling family oral fluids on the probability of PRRSV RNA detection by RT-rtPCR

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ABSTRACT

Family oral fluids (FOFs) are an aggregate sample type shown to be a cost-efficient and convenient option for determining the porcine reproductive and respiratory syndrome virus (PRRSV) status of weaning age pigs. This study investigates the effect of pooling PRRSV-positive FOF samples with PRRSV-negative FOF samples at different levels (1/3, 1/5, 1/10, 1/20) on the probability of PRRSV RNA detection by reverse-transcription real-time polymerase chain reaction (RT-rtPCR). Mathematical models were built to assess how much the probability of RT-rtPCR PRRSV detection changed with increasing proportion of PRRSV-positive samples present within pools and how partially sampling a farrowing room influenced the probability of RT-rtPCR detection of PRRSV RNA in pooled samples at different prevalence scenarios. A general example of a guideline for FOF-based sampling under different prevalence scenarios to detect PRRSV RNA by RT-rtPCR with at least 95 % certainty is presented. At the sample level, the probability of detecting PRRSV RNA by RT-rtPCR decreased from 100 % to 87 %, 68 %, and 26 % when diluting up to 1/20 for PRRSV positive FOF having an initial Cycle threshold (Ct) below 34, between 34 and 36, or above 36, respectively. When PRRSV prevalence is near-zero (1 or 2 litters positive out of 56), the most cost-efficient farrowing room sampling strategy to detect PRRSV RNA with at least 95 % certainty was pooling FOF samples up to 1/10; at higher prevalence ($\geq$ 3 of 56 litters positive), the most cost-efficient strategy was submitting samples in pools of 20. Subsampling a farrowing room for FOF pools was also demonstrated to be a valuable cost-saving strategy. Overall, based on the conditions of this study, pooling FOFs up to 1/20 is a valid option in situations of cost constraint and regardless of pooling level chosen, capturing as many litters as possible improves the probability of PRRSV detection.

1. Introduction

Porcine reproductive and respiratory syndrome virus (PRRSV) remains a prevalent pathogen in the global swine industry, causing substantial losses (Holtkamp et al., 2013; Nathues et al., 2017; Nieuwenhuis et al., 2012; Renken et al., 2021). PRRSV can sustain infections in swine breeding herds and suckling pig populations at very low prevalence (Linhares et al., 2014; Sanhueza et al., 2019; Trevisan et al., 2019a; Wills et al., 2003). Therefore, monitoring protocols in low prevalence scenarios require many pigs to be sampled to provide a high probability of virus detection. For example, to detect PRRSV at 3 % or 2 % prevalence, there is the need to sample 99 or 149 pigs respectively to provide a 95 % probability of detection of at least one positive sample when testing by reverse-transcription, real-time polymerase chain reaction (RT-rtPCR), assuming independence between samples, a confidence level of 95 % and a perfect test (Cannon and Roe, 1982).

Conventional PRRSV monitoring and surveillance systems include routine diagnostic testing for PRRSV nucleic acid (by PCR-based assays) and/or anti-PRRSV-antibodies (by serology) using biological samples, including oral fluids (Almeida et al., 2020; Prickett et al., 2008), processing fluids (Lopez, 2020; Trevisan et al., 2019a; Vilalta et al., 2018), or serum samples (Vilalta et al., 2018).

Oral fluids have been shown to be a valid sample type for PRRSV monitoring (Prickett et al., 2008; Prickett and Zimmerman, 2010) and are increasingly becoming the major swine sample type submitted to diagnostic laboratories in the US (Trevisan et al., 2019b). Recently,

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family oral fluids (FOF) sampling was described for monitoring PRRSV in breeding herds (Almeida et al., 2020) and is collected by exposing a dam and her suckling piglets to a rope for oral fluids collection. FOF are typically collected around the time of weaning (Almeida et al., 2020) and are used to ascertain the PRRSV status of due-to-wean pigs (Holtkamp et al., 2021).

Results from a recent work (Almeida, 2020) demonstrated that collecting 20 FOF per farrowing room provided an equivalent probability of detecting PRRSV as sampling 150 pigs (i.e., detecting virus infection at ~ 2 % prevalence). This has significantly improved monitoring and surveillance protocols but can be cost-prohibitive as many herds wean multiple farrowing rooms per week. For example, submitting 20 FOF samples per room from 2 rooms per week for RT-rtPCR testing ($25/sample) cost $1000/week. This has raised questions on the feasibility of pooling FOFs and consequent implications on PRRSV RNA detection, as has previously been done on some other population-based sample types (López et al., 2020; Rovira et al., 2007; Vilalta et al., 2019).

This work aims to assess the effect of the pooling FOF samples on the probability of PRRSV RNA detection by RT-rtPCR. Within this article, the effects of diluting 1-part PRRSV RNA positive FOF with 2 (1/3), 4 (1/5), 9 (1/10) and 19 (1/20) equal parts of PRRSV RNA negative samples on the results of the RT-rtPCR are described; and the effect of having multiple positive samples between or within pools on the probability of detecting PRRSV RNA in FOF by RT-rtPCR are assessed. The FOF sample size needed to select a PRRSV-positive pig litter at 95% confidence are also shown. Finally, the applicability of the built mathematical models is demonstrated by describing the minimum number of pools needed to detect PRRSV RNA with at least 95% confidence.

2. Materials and methods

2.1. Family oral fluids collection

Each FOF sample was obtained by tying a cotton rope to the side rail of farrowing crates to about the joint height of the suckling piglets and allowing the dam and piglets to chew on it (Almeida et al., 2020). The oral fluids content on the cotton rope were then wrung out into a clean disposable plastic bag and transferred to a sterile 50 ml conical centrifuge plastic tube (Fisher Scientific, Pittsburgh, PA, USA).

2.2. Sample preparation and dilution of PRRSV-positive samples with PRRSV-negative ones

A total of 713 and 16 FOF samples were obtained from PRRSV endemic and PRRSV naïve breeding herds, respectively, between October 2019 and November 2020.

The PRRSV-negative FOF samples collected from PRRSV-negative breeding herds were confirmed to be PRRSV RT-rtPCR negative and thereafter pooled, vortexed, and divided into new 50 ml conical centrifuge plastic tubes (Fisher Scientific, Pittsburgh, PA, USA). Eligible PRRSV-positive samples had at least 2.7 ml to accommodate dilution volumes. Each sample was vortexed for one minute before distribution into the 5 ml tubes in volumes shown in Table 1.

Subsequently, these were vortexed and distributed in 6 replicates (240 µl per replicate) each into dummy plates (Costar® Assay Plate, Corning Inc., Corning, NY, USA). Each plate contained positive (n = 1) and negative (n = 1) controls and all dilution levels of 3 samples. The dummy plates were then stored for up to 2 h at 4 °C before proper plating for extraction (200 µl per replicate). All samples were tested at [information redacted as requested by Preventive Veterinary Medicine] for PRRSV-2 RNA by RT-rtPCR to confirm PRRSV status and establish Ct values.

Table 1
Volumes of PRRSV-positive FOF samples and PRRSV-negative FOF samples (diluent) per dilution level.

Dilution level	PRRSV-Positive samples: Diluent (Volume in µl)
Undiluted stock solution	1650 : 0
1/3	550 : 1100
1/5	330 : 1320
1/10	165 : 1485
1/20	82.5 : 1567.5

2.3. PRRSV RNA extraction and RT-rtPCR

Nucleic acids were extracted from the samples using the RealPCR* DNA/RNA Magnetic Bead Kit (IDEXX Laboratories, Inc., Westbrook, ME, USA) as described by (López et al., 2021). RT-rtPCR was then performed on the extracted nucleic acids using the RealPCR* PRRSV-1 and PRRSV-2 Multiplex RNA Mix and Master Mixes (IDEXX Laboratories, Inc., Westbrook, ME, USA) as previously described (Lopez, 2020). The resulting standard curves after amplification were analyzed using automated functions of the RT-rtPCR machine. Ct values below 40 were considered PRRSV positive; any Ct values above 40 were considered negative.

2.4. Statistical analyses and modeling

Eligible samples were categorized into three groups based on the mean Ct values of their respective undiluted solutions. The Ct values below 34 ($n = 8$), between 34 and 36 ($n = 8$), or higher than 36 ($n = 9$) were labeled A, B, and C, respectively. Each sample had five dilution levels having six replicates each, totaling 30 observations per sample.

2.5. Effect of dilution on sample Ct values

A linear mixed regression model was used to characterize average Ct value changes across dilution levels of the eligible samples. In this model, the Ct value was the dependent variable, dilution level was the fixed effect, and sample identification number was used as a random effect. A plot was created to describe Ct distribution per dilution for each of the three categories (Fig. 1). These analyses were performed using the library lme4 (Bates et al., 2014) in R program (R Core Team, 2019).

2.6. Probability of PRRSV RNA detection based on laboratory results (1 positive FOF sample within negative pools)

A probit regression model using the brglm function (Bias Reduction in Binomial-Response Generalized Linear Models) estimated the probability of detection (*Ppcr*) for each category with a binary outcome (RT-rtPCR positive or negative) as the response variable and dilution level as

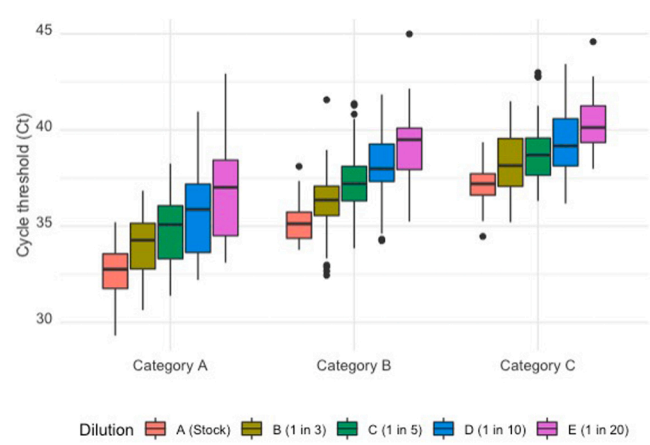

Fig. 1. PRRSV RT-rtPCR Ct changes per dilution level for each of the 3 Ct value categories.

the predictor (categorical) variable:

$$\Phi^{-1}(\widehat{P}pcr) = \widehat{a} + \widehat{b}x \quad (1)$$

where Φ^{-1} is the probit link function, a is the intercept, and b is the regression coefficient associated with the dilution (x). Least-square means for each level of the predictor were compared using a t-test with Tukey-Kramer adjustment, and P-values < 0.05 were considered significant. This analysis was performed on the brglm package (Kosmidis and Firth, 2021) in R program (R Core Team, 2019).

2.7. Probability of PRRSV RNA detection by PCR with more than 1 positive FOF in pools

The probabilities of detection (Pd) for different numbers of positive FOFs within pools was calculated by combinatory analysis assessing all possible sample arrangements (i) in which at least one positive FOF would be included in any pool (p_i). Subsequently, for each (P_i), the dilution (d_i) was calculated based on the number of positive and negative FOFs and the probability of a positive RT-rtPCR $\widehat{P}pcr_i$ estimated using the probit model fitted previously in Eq. 1. Finally, Pd equals the probability of a union of events of the product of $\widehat{P}pcr_i$ and P_i:

$$Pd = \sum_{i=1}^{n} \widehat{P}pcr_i * P_i \quad (2)$$

where n is the n-*th* sample arrangement with at least one positive FOF. Note that the possible arrangement are disjoint events and the probability of a union of events reduces to the sum. For more details about P_i calculation, see Supplementary MaterialI.

2.8. Probability of sampling at least one PRRSV-positive litter at ≥ 95% confidence given different prevalence scenarios, and implications on PCR detection

To assess the FOF sample size required to select at least 1 PRRSV-positive litter with at least 95% certainty in at least one pool, the following assumptions were made:

i) PRRSV-positive litters are randomly distributed in the farrowing room;
ii) all individual PRRSV-positive FOF samples within each pooling scenario were within the same Ct category;
iii) all the combinations for which a positive FOF sample could be included in any pool had equal chances of occurring; and iv) the RT-rtPCR test was perfect (100% sensitivity, 100% specificity).

The probability of sampling at least one positive crate (Pc) equals 1 minus the specific case of a hypergeometrical mass function where the number of successes is zero:

$$Pc = 1 - \frac{C_T^{n-k}}{C_T^n}, \quad (3)$$

where n and k are the total and the positive number of crates in the room, respectively; T is the number of FOFs sampled.

The minimum number of T that provided Pc greater than or equal to 95% results from

$$T_{95} = \left[1 - (0.05)^{\frac{1}{k}}\right] * \left[n - (\frac{k-1}{2})\right]. \quad (4)$$

The values were identified for future reference regarding sample size calculations (Table 2).

The probability of PRRSV RNA detection will be affected if not all the crates in a room are sampled. Thus, it is assumed that the event "detection of PRRSV RNA" results from sampling a positive litter and having a positive RT-rtRCR result from at least one pool. Part of this expression is already described in Eq. 2, and the hypergeometrical mass function describes the final probability of detection of x positive FOFs:

$$P(X = x) = \frac{C_x^k C_{T-x}^{n-k}}{C_T^n} \quad (5)$$

and multiplied by the expression inside the summation of Eq. 2:

$$\widehat{P}pcr_i * P_i * P(X = x). \quad (6)$$

The final probability is the summation over all I arrangements of pools with positive FOFs. The numeric values for the probability of PRRSV RNA detection in different scenarios are available in Supplementary Material I Table S1.

Table 2
Number of FOF pools (and crates) required to be at least 95% certain that ≥ 1 positive FOF sample is selected in a 56-crate room, given an assumed numbers of positive crates in that room.

Number of Positive FOF[a]	Pool size			
	3 FOFs[a]	5 FOFs[a]	10 FOFs[a]	20 FOFs[a]
4	10 (30 crates)	6 (30 crates)	3 (30 crates)	2 (40 crates)
3	12 (36 crates)	7 (35 crates)	4 (40 crates)	2 (40 crates)
2	15 (45 crates)	9 (45 crates)	5 (50 crates)	2.8 (56 crates)
1	18 (54 crates)	11 (55 crates)	10.6 (56 crates)	2.8 (56 crates)

[a] FOF= Family Oral Fluids

2.9. Simulation describing the minimum number of pools needed to detect PRRSV RNA by RT-rtPCR with at least 95 % certainty in a 56-crate room

Using the mathematical model described in Eq. 6, pool sizes of FOFs required to have at least 1 RT-rtPCR positive test from a 56-crate farrowing room were calculated. For this simulation, the Pd for all three Ct categories for each scenario of estimated positive crates was averaged. This average was weighted based on the Ct distribution of all PRRSV positive FOF samples that were obtained from field sampling. Out of all 137 PRRSV positive samples obtained from the field, 109 were in Ct category A, 21 were in Ct category B, and 7 were in Ct category C. The minimum number of FOF pools required to give an average Pd of at least 95% was obtained.

3. Results

Five out of the thirty tested PRRSV-positive FOF samples were excluded from the statistical analysis as their undiluted replicas' mean (Ct) values were ≥ 40 (i.e., negative). There was an average increase of 1.28, 2.09, 2.90, and 4.12 in Ct values of FOF stock samples when diluting at 1/3, 1/5, 1/10, and 1/20 levels, respectively, across the Ct categories (Fig. 1).

3.1. Probability of PRRSV RNA detection in FOF samples, based on RT-rtPCR results

In general, there was no difference in the probability of PRRSV RNA detection in samples diluted 1/3 compared to undiluted, regardless of the Ct category. Overall, there were decreases in the probability of PRRSV RNA detection as the dilution factor increased from 1/3–1/20 (Table 3).

The least-square mean for the detection probability for the stock solution in Ct category A (< 34) was not significantly different ($P \geq 0.23$) from all dilution levels. The least-square mean for the detection probability for the stock solution in Ct category B (34 ≤ Ct ≤

Table 3
Probabilities of detection for five different pooling levels (dilutions), according to the cycle threshold (Ct) categories A, B, and C.

Pooling level	Mean (95% CI)		
	Category A: Ct < 34	Category B: 34 ≤ Ct ≤ 36	Category C: Ct > 36
No dilution	0.99[a] (0.82, 1.00)	0.99[a] (0.82, 1.00)	0.99[a] (0.84, 1.00)
1/3	0.99[a] (0.82, 1.00)	0.97[a] (0.83, 1.00)	0.79[ab] (0.63, 0.91)
1/5	0.99[a] (0.82, 1.00)	0.93[a] (0.78, 0.99)	0.65[bc] (0.47, 0.80)
1/10	0.97[a] (0.83, 1.00)	0.91[ab] (0.75, 0.98)	0.54[c] (0.36, 0.70)
1/20	0.87[a] (0.70, 0.96)	0.68[b] (57.1, 80.4)	0.26[d] (0.13, 0.44)

[a,b,c,d]Means within a column with different superscripts are statistically different ($P \leq 0.05$). Least-square means for each dilution level were compared using a t-test with Tukey-Kramer adjustment for pairwise comparisons.
[1]SEM = standard error of the mean.

36) was not significantly different ($P \geq 0.43$) from the detection probability for the 1/3, 1/5, and 1/10 dilution levels but significantly different for the 1/20 dilution level ($P = 0.01$). The least-square mean detection probability for the undiluted stock solution in Ct category C (Ct > 36) was not significantly different ($P = 0.05$) from the detection probability for the 1/3 dilution level but significantly different ($P < 0.01$) from all the other dilution levels.

3.2. *Estimating the probability of PRRSV RNA detection by RT-rtPCR when two or more positive FOF samples are present within and/or between pools*

The probability of detection increased with increasing number of positives within submitted FOF pools. Pools of 20 FOFs had a visibly low probability of detection if all RT-rtPCR positive FOFs within pools were in category C. Overall, all pools with positive samples within categories A and B had relatively high detection probabilities (Supplemental Material Table S1; Figs. 2 and 3).

3.3. *Estimating the probability of PRRSV RNA detection by Rt-rtPCR at different prevalence scenarios given that a farrowing room is sampled at ≥ 95% confidence*

When positive FOFs have Ct within categories A or B, the combined probability of sampling at least one PRRSV positive litter/FOF into any pool and detecting at least one pool by RT-rtPCR is high. It gets close to 100% as the number of positive litters in the room increases. Detection probabilities for pools of 10 and 20 when positive FOFs are within

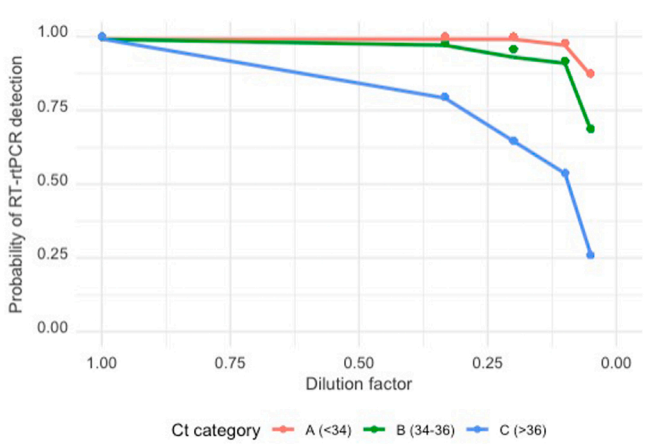

Fig. 2. Probability of PRRSV RNA detection by dilution factor for each Cycle threshold (Ct) category. The dilution factors 1, 0.33, 0.2, 0.1, and 0.05 represent undiluted, PRRSV RNA-positive FOF pooled in 1/3, 1/5, 1/10, and 1/20 with RT-rtPCR-negative FOF, respectively. The data points represent the mean detection probabilities for each dilution level for each of the Ct categories.

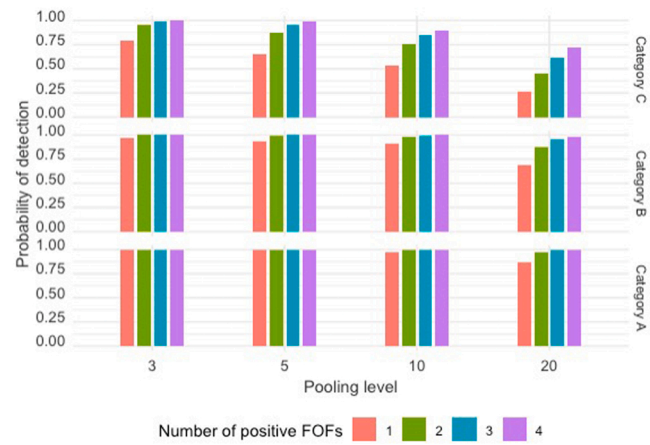

Fig. 3. Probability of detecting PRRSV RNA by RT-rtPCR in FOF pools within each Cycle threshold (Ct) category by the number of PRRSV-positive FOF samples within pools (n = 1–4). Category A pools have positive FOFs with Ct values less than 34, Category B pools have positive FOFs with Ct values between 34 and 36, Category C pools have positive FOFs with Ct values higher than 36.

category C are relatively low. (Supplemental Material I Table S2; Fig. 4).

According to the simulation, increases in the litter prevalence would generally imply fewer FOFs to be sampled and pooled in order to detect PRRSV with at least 95 % certainty. When there is only one positive litter in a 56-crate room (i.e., 1.79 % -litter prevalence), If the sampling decision is to submit pools containing three individual FOFs each, eighteen such pools will have to be collected to have at least 95 % certainty of having a positive RT-rtPCR test. On the other hand, only three such pools are required when the -litter prevalence increased to 26.79 % (Table 4).

4. Discussion

The overall objective of this study was to assess the effect of pooling family oral fluids on the probability of detecting PRRSV RNA by RT-rtPCR. Better understanding the implications of pooling FOF samples facilitates more informed strategic decisions on sampling and pooling for PRRSV monitoring programs.

A set of mathematical and statistical methods for different outcomes were utilized in this study. Firstly, the effect of diluting positive samples on RT-rtPCR cycle threshold values was described. The changes in the

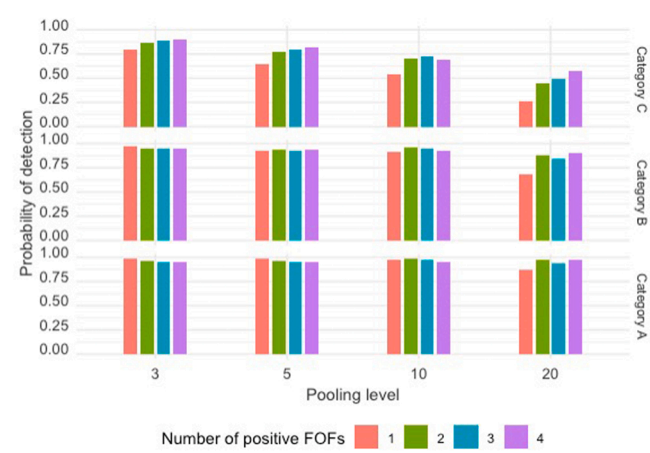

Fig. 4. Changes in probability of PRRSV RNA detection by RT-rtPCR in at least one sample, with increasing number of PRRSV-positive litters for each Cycle threshold (Ct) Category in a sampled 56-crate room. Category A pools have positive FOFs with Ct values less than 34, Category B pools have positive FOFs with Ct values between 34 and 36, and Category C pools have positive FOFs with Ct values higher than 36.

Table 4
Simulation describing the minimum number of pools needed to detect PRRSV RNA by PCR with at least 95% certainty in a 56-crate room.

PRRSV-positive litters	Litter prevalence[a]	Level of pooling[b]			
		3	5	10	20
1	1.79 %	18	11	6[c]	NP[d]
2	3.57 %	15	10	5	NP[d]
3	5.36 %	12	8	4	3[c]
4	7.14 %	10	7	4	3[c]
5	8.93 %	9	6	3	2
6	10.71 %	7	5	3	2
7	12.50 %	7	4	3	2
8	14.29 %	6	4	2	2
9	16.07 %	6	4	2	2
10	17.86 %	5	3	2	2
11	19.64 %	5	3	2	1
12	21.43 %	4	3	2	1
13	23.21 %	4	3	2	1
14	25.00 %	4	3	2	1
15	26.79 %	3	2	2	1

[a] (Number of positive litters/Total number of litters).
[b] Number of individual FOF samples per pool.
[c] One pool will have fewer FOFs than the others.
[d] NP = Not possible.

probability of PRRSV RNA detection with increasing proportions of positive FOFs within pools were then characterized, extending this concept to situations where not all crates or litters in a farrowing room are sampled. The mathematical expression (formula) for the minimum sample size required to detect at least one PRRSV-positive litter at 95% confidence given different prevalence scenarios was thereafter provided. Finally, the minimum number of FOF pools needed to detect PRRSV RNA by RT-rtPCR with at least 95% certainty in a 56-crate room was provided. The summary for the mathematical and statistical techniques used is in Fig. S1.

Ideally, Eq. (1) would be a mixed probit model with individual samples used as random effects; it is however not possible to include random effects on models penalized to adjust for data separation (data separation occurred because all undiluted replicates across all samples were always PRRSV RT-rtPCR positive). Regular logistic regression models that allow for random effects will fail to converge if data is separated (Allison, 2008), so a bias reduction model was used instead. (Kosmidis and Firth, 2021). PRRSV negative FOFs used in this study were obtained from PRRS naive herds as opposed to what is obtainable in a real-world scenario where all FOF samples within pools are from the same herd; Rt-rtPCR-negative FOF samples from PRRSV-positive herds may still contain PRRSV RNA below the analytic sensitivity threshold of a RT-rtPCR test (Wagstrom et al., 2000). We expect that the probability of RT-rtPCR detection of PRRSV positive FOFs diluted with PRRSV negative FOFs from PRRSV endemic herds will be better than what was observed in this study.

Previous studies agree that pooled samples are suitable for ascertaining the herd status of an infection (Christensen and Gardner, 2000) and are useful in low prevalence conditions, especially when pooling does not significantly affect test sensitivity (Cowling et al., 1999). Rotolo et al. (2017) recommended testing multiple rooms in a barn to truly assess herd status for PRRSV at any given time point; the option of pooling is expected to make this exercise less cost-prohibitive, given the importance of effective monitoring in PRRSV control schemes (Pileri and Mateu, 2016).

Calculating sample sizes at 95 % confidence (for the purpose of pooling) was explored using simple combination functions under the assumptions earlier listed; thereafter, separating the "at least 1 positive" probabilities used to calculate sample sizes for the pools into the exact component probabilities. The probability for PRRSV RNA detection by RT-rtPCR was then calculated for the component probabilities. The final probability for RT-rtPCR detection across all component probabilities for each of the sample sizes was then calculated. The essence of this exercise was to demonstrate how omissions within the 5 % error could influence the probability of PRRSV RNA RT-rtPCR detection for pooled samples obtained from rooms subsampled in a simple random pattern.

Fifty-six was chosen as the number of crates in a farrowing room in this study's simulation for the purpose of practicality; some relatively large commercial farrowing rooms have exactly 56 farrowing crates. Secondly, this room size is relatively easier to use to simulate low litter prevalence. For instance, having two PRRSV-positive litters in a 56-crate room would mean a lower PRRSV prevalence than having two positives in a 24-crate room.

As seen from these results, the Ct of a positive sample within a pool of negatives influences the probability of RT-rtPCR detection of that pooled sample. To the best of the authors' knowledge, there has not been any published study relating Cts of FOF samples to any predictors. There has however been a report, for herds that adopted measures to eliminate PRRSV, of increasing RT-rtPCR Ct values over time in processing fluids samples post-PRRSV infection (Trevisan et al., 2019a). One could also make informed guesses as to the possible Ct ranges of FOFs based on other factors such as: recent diagnostic test results from the herd, deficits in productivity parameters, etcetera. Data from the current study suggests Cts from FOFs, regardless of PRRSV prevalence, are mostly within categories A and B.

Data obtained from the Swine Disease Reporting System, a data aggregation project of multiple Veterinary Diagnostic laboratories in the United States (www.fieldepi.org/SDRS) indicated that for suckling pigs' oral fluid samples tested for PRRSV RNA by RT-rtPCR for the year 2020, a total of 154 submissions had at least 1 sample testing positive. Considering all 154 submissions, the Ct values of all positive samples had a mean minimum Ct of 30.14 ± 2.70, a mean average Ct of 31.48 ± 2.39 and a mean maximum Ct of 32.94 ± 2.80. There is no information on the proportion of these submissions that were FOFs, compared to litter oral fluids (defined as oral fluids from suckling piglets not including respective dams). This further supports that Ct from PRRSV-positive oral fluids obtained from piglets are mostly within Ct category A; consequently, pooling up to 20 FOFs will not significantly reduce the probability of PRRSV RNA detection by RT-rtPCR.

Previous studies had expressed concerns over getting false negatives after pooling oral fluids and recommend testing fewer individual samples rather than pooling when there are cost constraints (Henao-Diaz et al., 2020). These deductions were however not from experiments done on FOFs which would have relatively fewer pigs contributing to a sample compared to typical pen oral fluids. From the presented simulations, the importance of sampling more crates is evident; contingent upon pooling level and number of PRRSV-positive litters present, significant sensitivity is not lost. One demerit with capturing a small fraction of a herd within a submission of FOFs to be tested, especially in low prevalence conditions, is the significant probability of not capturing the positive animal or litter within such submissions. When no positive animal/litter in a herd is captured in a FOF submission, the probability of detecting a PRRSV positive room/herd by RT-rtPCR reduces to 0. When there is cost constraint for monitoring, especially in low prevalence, pooling FOFs to capture as many litters as possible between pools is a favorable option.

One common concern with FOF is that the litter members must be willing to interact with the sampling ropes. As has been previously demonstrated (Almeida et al., 2020; Boulbria et al., 2020), it is not uncommon to find litters within a room not effectively interacting with the sampling ropes. An effective rope interaction is defined here as interaction between a litter and a sampling rope that provides sufficient volume of oral fluids for collection and storage. As with conventional population sampling to detect a pathogen, the goal is not necessarily to sample every unit (this would be ideal), but to sample a minimum number of observational units to capture at least one positive sample at a given prevalence estimate under a pre-defined confidence level (usually 95 %), taking into account the diagnostic test sensitivity and specificity

(Fenn Buderer, 1996).

Generally, pooling has the advantage of capturing a sizeable number of animals within one sample, making it possible to test more animals within a single sample (Christensen and Gardner, 2000). On the other hand, pooling can lead to disadvantages, for example, a positive sample (s) within a pool can be diluted beyond test detection (Christensen and Gardner, 2000). Also, there may be a need to retest each component sample in a pool to determine the sample(s) that is/are positive.

PRRSV distribution in pens have been shown to be more geospatial than random (Almeida, 2020; Rotolo et al., 2017) The simulations did not adjust for diagnostic test accuracy (it was assumed the RT-rtPCR tests have 100 % sensitivity and specificity).

This study estimated prevalence for sampling by considering each litter as a unit as opposed to the more conventional methods that estimate prevalence by considering each individual pig a unit. It may not be practical to have exactly equal volumes of component FOFs within a pool during field sampling. Component FOFs within a pool may have different Cts. The numbers calculated in the simulation here presented are based on a 56-crate room; these could change with a different room size.

5. Conclusions

The use of pooled family oral fluids to monitor breeding herds and/or determine PRRSV status of weaning age pigs is a valuable option. Probability of RT-rtPCR detection of PRRSV RNA decreased with increasing dilution and increasing Ct value of a positive PRRSV FOF sample. Litters can be viewed as discrete sampling units and conventional sample size related calculations can be applied e.g., sampling at $\geq$ 95 % confidence.

For the simulation here created, pooling up to 10 and 20 FOFs provided the best cost-saving strategy when inter-litter prevalence was less than 4 % and more than 5 % respectively, at 95 % confidence of detection. Especially in low prevalence, it is advisable to capture as many litters as possible with the least pooling one's budget can support. It is hoped that, amongst other things, the simulations performed here would provide additional information to the on-going development of sampling guidelines for oral fluids use in swine medicine.

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Conflict of interest

The authors of this work declare no conflict of interest.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prevetmed.2022.105701.

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